Machining Processes:
Drilling

INDUSTRIAL PROCESS
related to machining

Groover M.P.: Chapters 18 and 19

Learning Targets

The student knows:

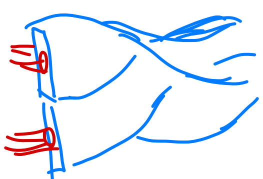
- Basic principles, tools and machine tools;
- How to evaluate cutting force, power and work;
- How to define the appropriate process parameters.

PROCESS OVERVIEW

HOLE MAKING → produces holes in material → DRILLING ≡ most common hole making by linear feed + relative rotation

sym holes

DRILL TIP



LUBRICANT

for Deep hole it is difficult the chip control...

we helps with lubricant to remove chips (COOLANT)

NOT RIGID tool...   Rigidity $\propto 1/L^4$

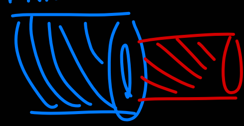
there are multiple issues in DRILLING

Drills { short hole → small depth respect diameter
Deep hole → challenging!

↓ special DRILLS, GUIDEWAY required

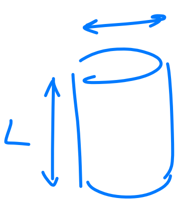
TREPANING by removing a periferical cylinder

DRILL



removed autonomous → large diameters holes requiring less power

D



L

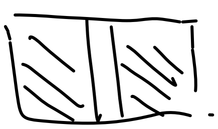
$l < 5 \div 6\ D$

$l < 2.5\ D$ if big diameter

long dwell and frequent chip removal

PECK DRILLING

Through hole



Blind hole

interrupt hole

Many different solutions to make holes!

HOLE MAKING: different kind of holes...

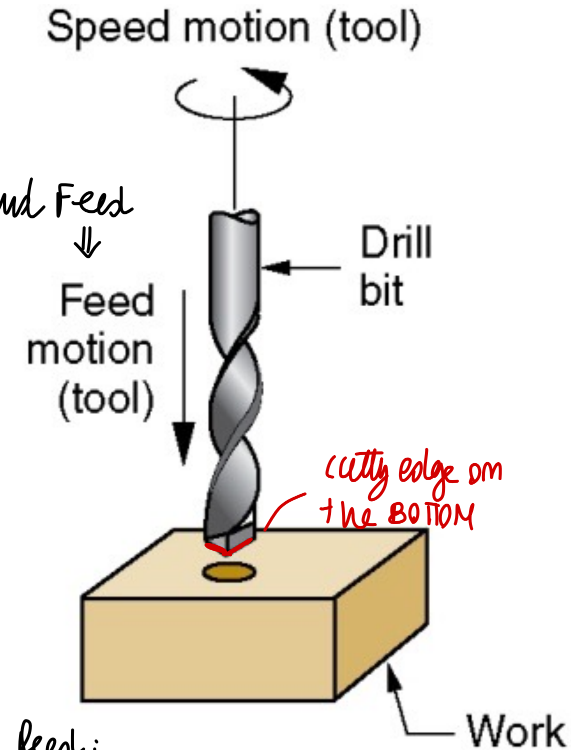
Creates a round hole in a workpiece

Cutting tool called a *drill* or *drill bit*

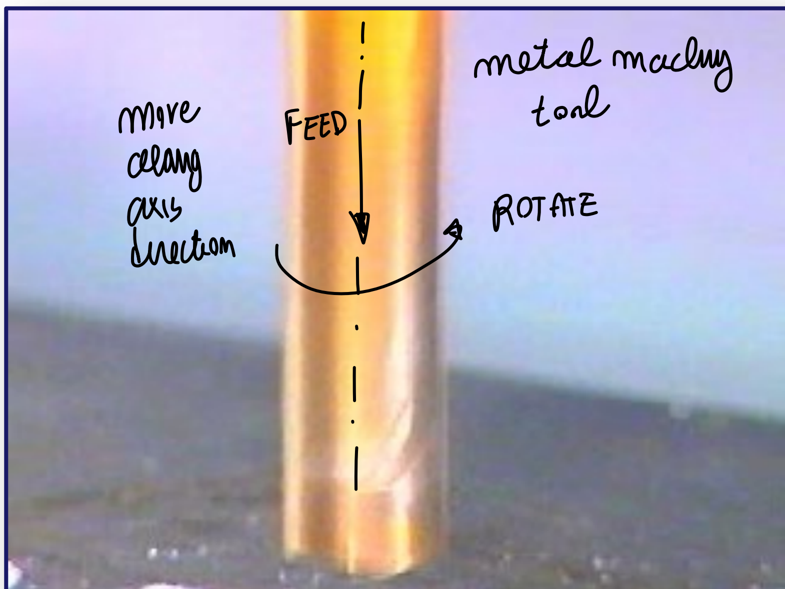
Machine tool: *drill press*

We need CUTTING MOTION

and Feed
⇓



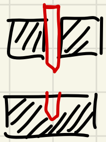
tool axis = hole axis



easy feed:
move the tool
on direction of the hole ... well
defined direction of motion

enter the part, than remove the chips by rotating in opposite side for machining → you make holes in the material

- through hole
- blind hole



We can obtain complex shape holes (cylindrical, ecc.)

We enter the hole, machining the part (removing material) and exit the hole with opposite motion respect original one

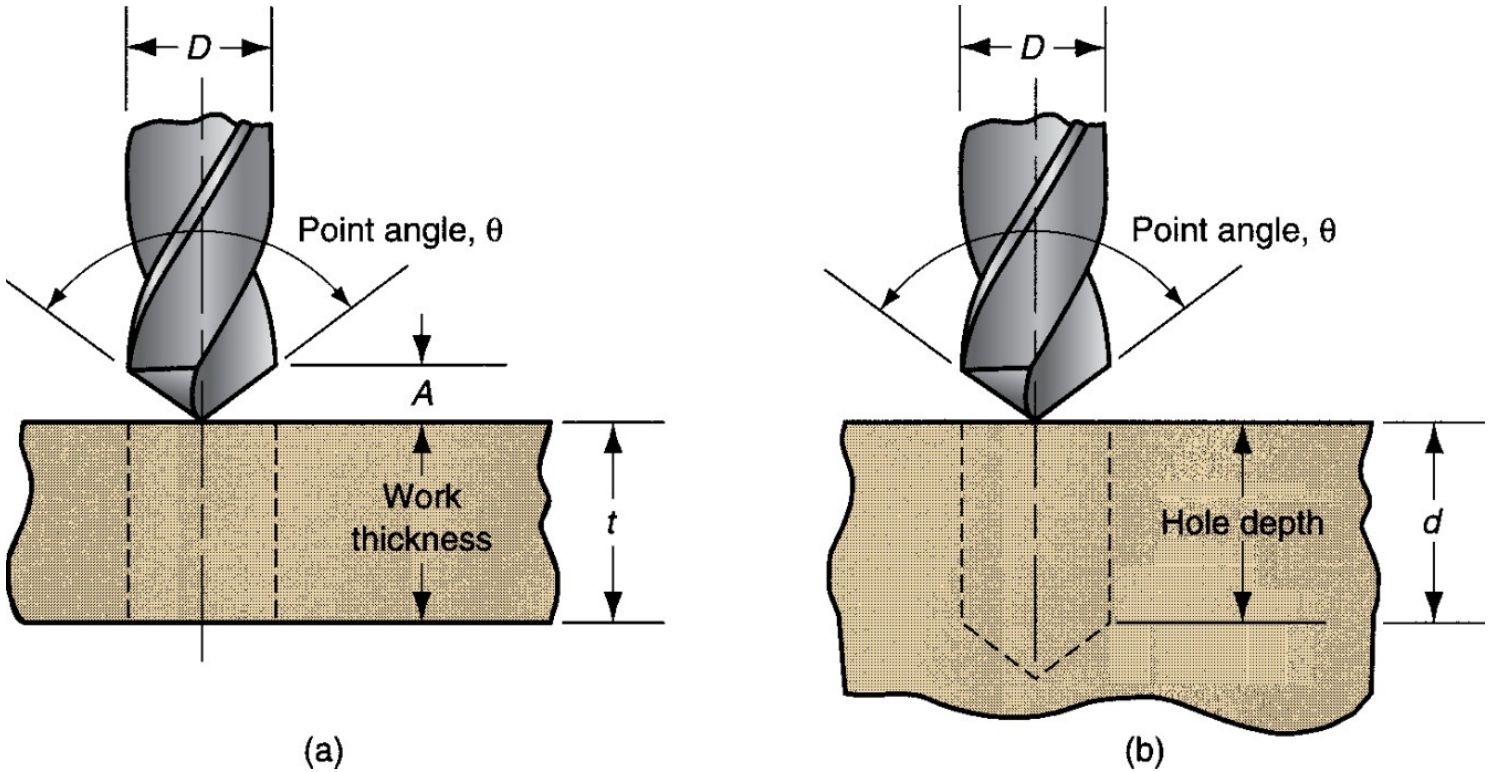
during exiting we JUST enable chips to go outside

↓↑ enter-exit cycle to remove easily the material and make the hole (easy chip removal)

Through Hole vs. Blind Hole

(a) Through hole - drill exits opposite side of workpiece

(b) Blind hole – drill does not exit opposite side



Hole Types

HOLE MAKING (different possible geometries of the hole)

different possible holes feature (with one or multiple tool) to perform different geometries hole

A) Through

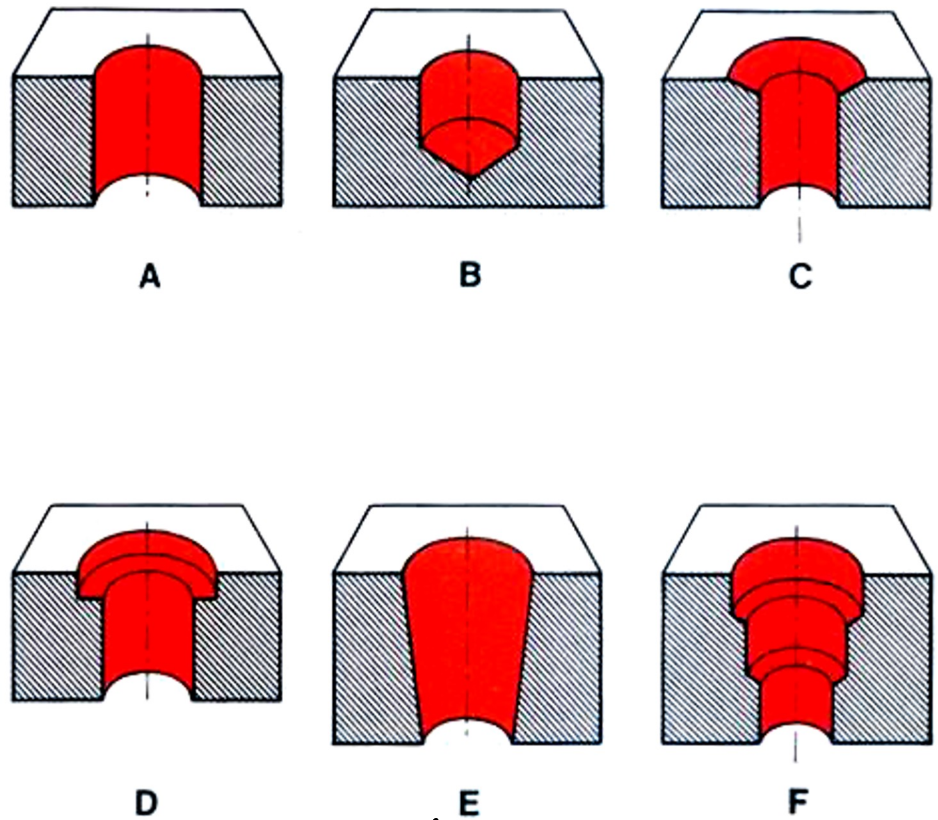
B) Blind

C) Countersink

D) Counterbore

E) Conical

F) Multiple feature

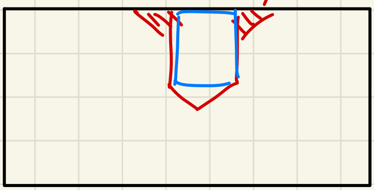


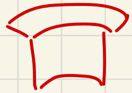
conical tool needed (NON CYLINDRICAL tool)

MULTI CYLINDRICAL holes

C) COUNTERSINK

Cylindrical hole + conical initial shape to insert an element inside the hole in an easy way

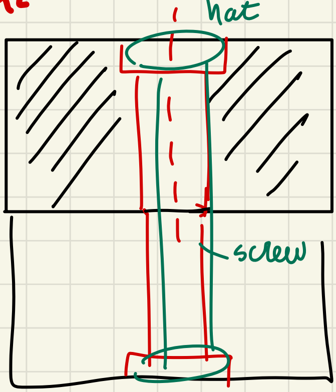




this conical enter enable to easy insert a piece

D) COUNTERBORE

So position the head and screw respect the part



Way to position two elements by a screw

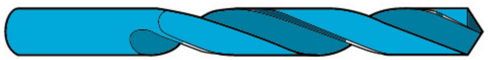
top cylinder bigger, to let mechanical element to JOIN toghether

OTHER TYPES OF DRILLS

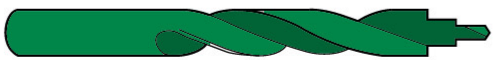
different kind of tools to obtain different HOLE FEATURES

with objective: remove material with feed motion // axis + ROTORY MOTION = CUTTING

(a) Twist drill
CLASSILAC
Really lot Used



(b) Step drill

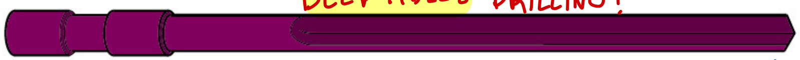


(c) Straight-flute drill

(d) Spade drill

general holes
$L \leq 5D$

(e) Gun drill JUST one cutting edge
DEEP HOLES DRILLING!

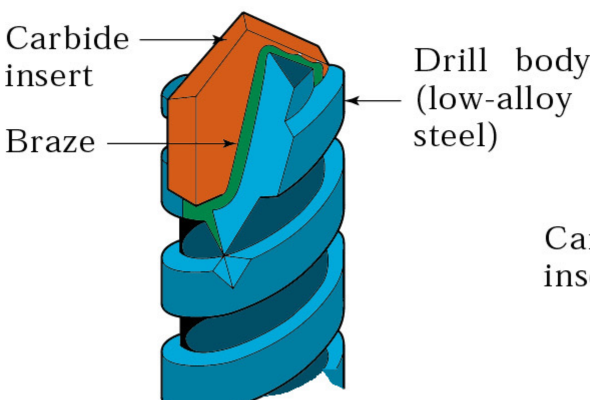


BIG HOLES
$L \leq 2.5D$

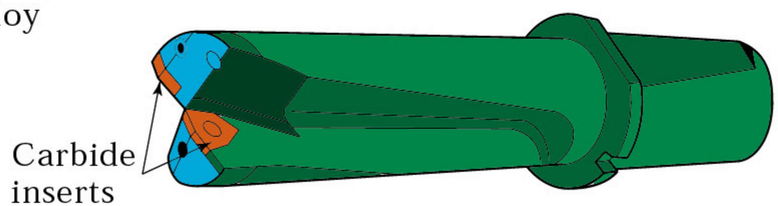
NOT a twist geometry, instead you have a continuous drille → usefull for deep holes

$\leq 100D$

(f) Drill with brazed carbide tip



(g) Drill with indexable carbide inserts



INSERTS mounted on the head

- Big holes: $L \leq 2.5D$ → $5 \cdot D < L \leq 100D$ Deep holes

- General holes $L \leq 5D$ usually IF

⇓ talking about the BENDING! (considering the CROSS SECTION OF the tool)

INERTIA

$$J = \frac{\pi D^4}{64}$$ small value (if D small)

IF L is relevant we will have PROBLEMS!

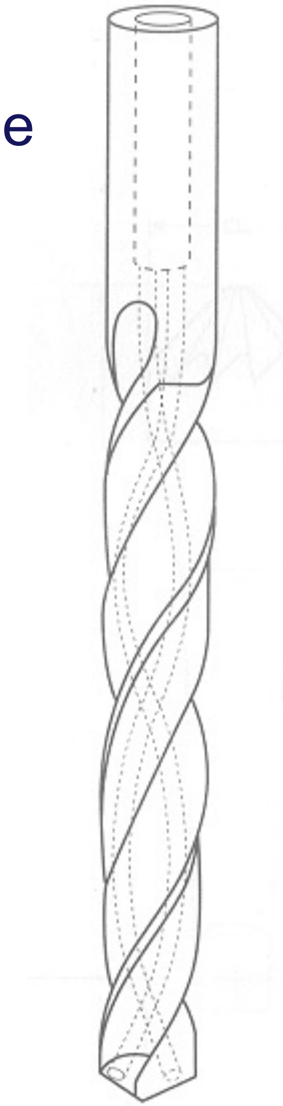
J small? you need special machines to perform deep hole drilling

TOOL: TWIST DRILL

(Basic tool)

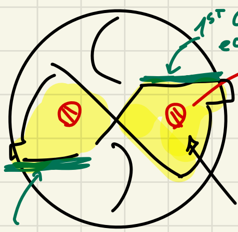
The helicoidal flutes allow the chip removal.

If the drill has a big diameter, it can be hollow to allow the passage of the cutting fluid to the cutting zone.



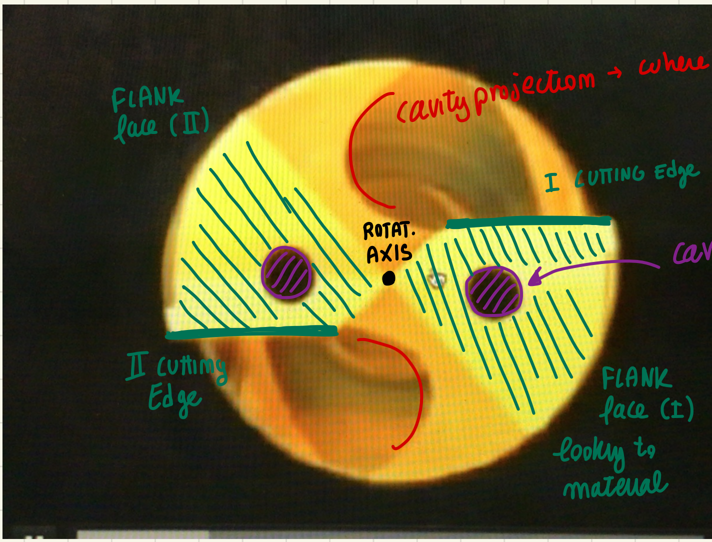
{ BOTTOM VIEW }

(view from below)



TWIST DRILL

Bottom view of the TOOL



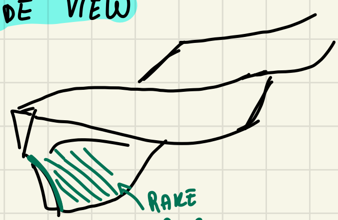
cavity rotating around the ROTATIONAL axis

two flutes...

space left to remove CHIP from the HOLE

we need space for chip removal

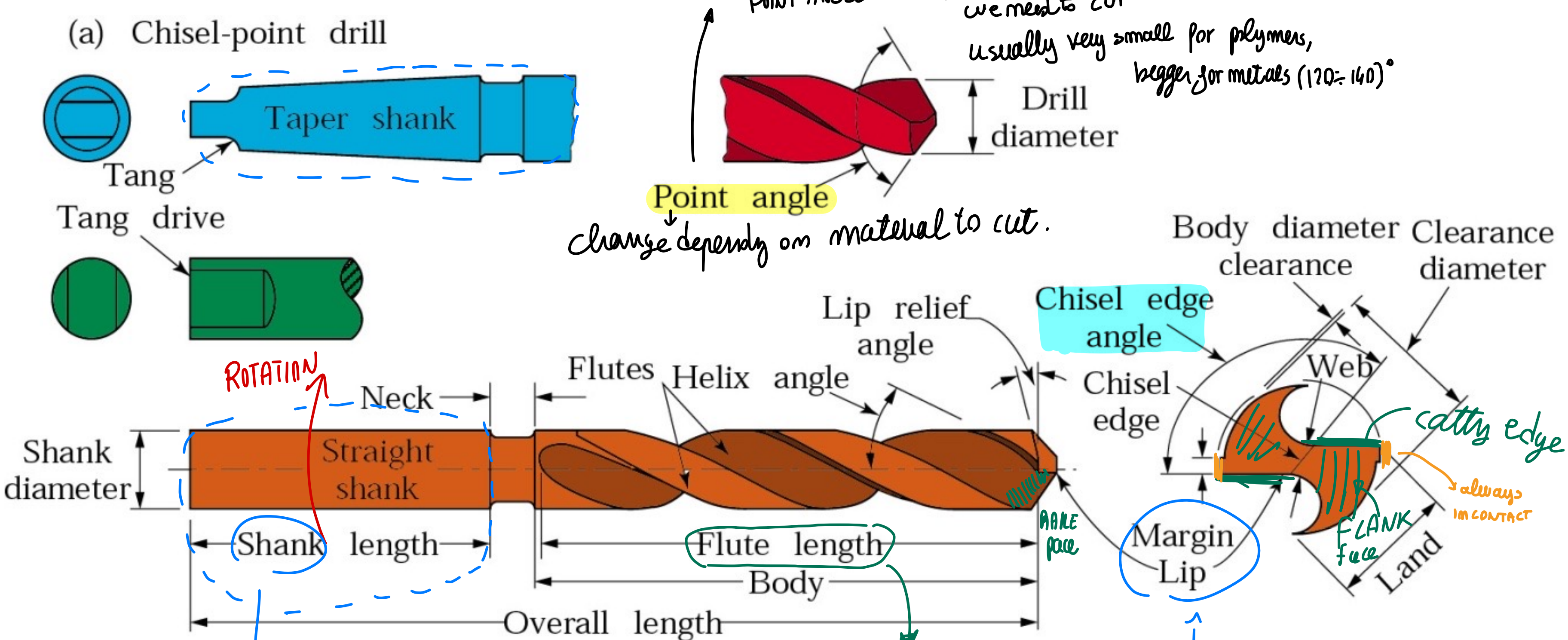
SIDE VIEW



Rotating tool → cutting motion tangent to ROTATION ⇒ feed along axis

TOOL: TWIST DRILL

Created by the cutting edges...
POINT ANGLE changes respect the material we need to cut
usually very small for polymers, bigger for metals (120÷140)°



Point angle ↓ change depending on material to cut.

TOOL BODY
⇓
enabling the connection of the TOOL to the spindle of the machine

shank transmit torque to the Body of the tool through the SHANK -- connected to SPINDLE

portion of body with HELIX HOLE

POSSIBLE hole dimension depends on the FLUTE length

elements done to reduce contact and have less friction

Tool: Twist Drill

IT10 tollerance size requirements on tool! → we can reach IT7 in good case!

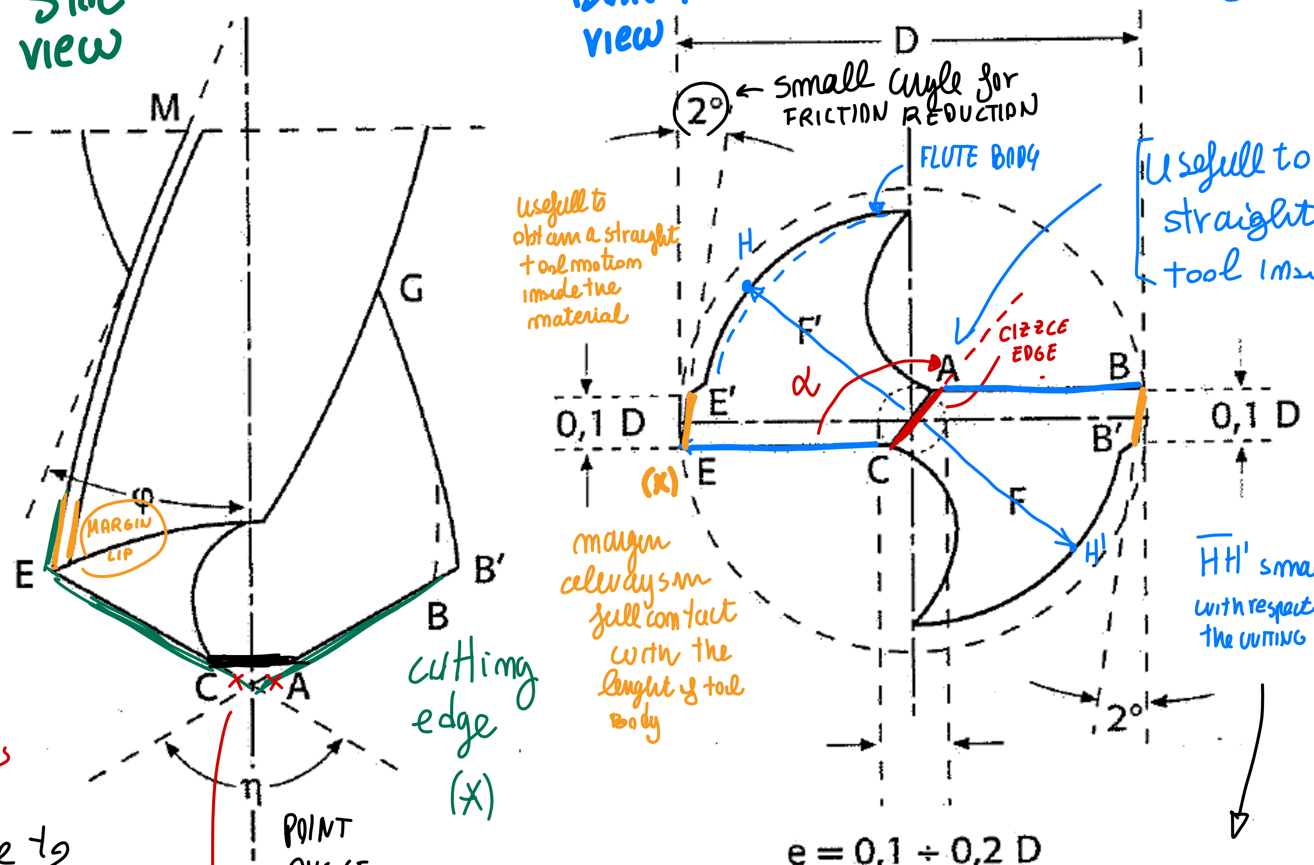
Side view

Bottom view

AB, CE cutting edges

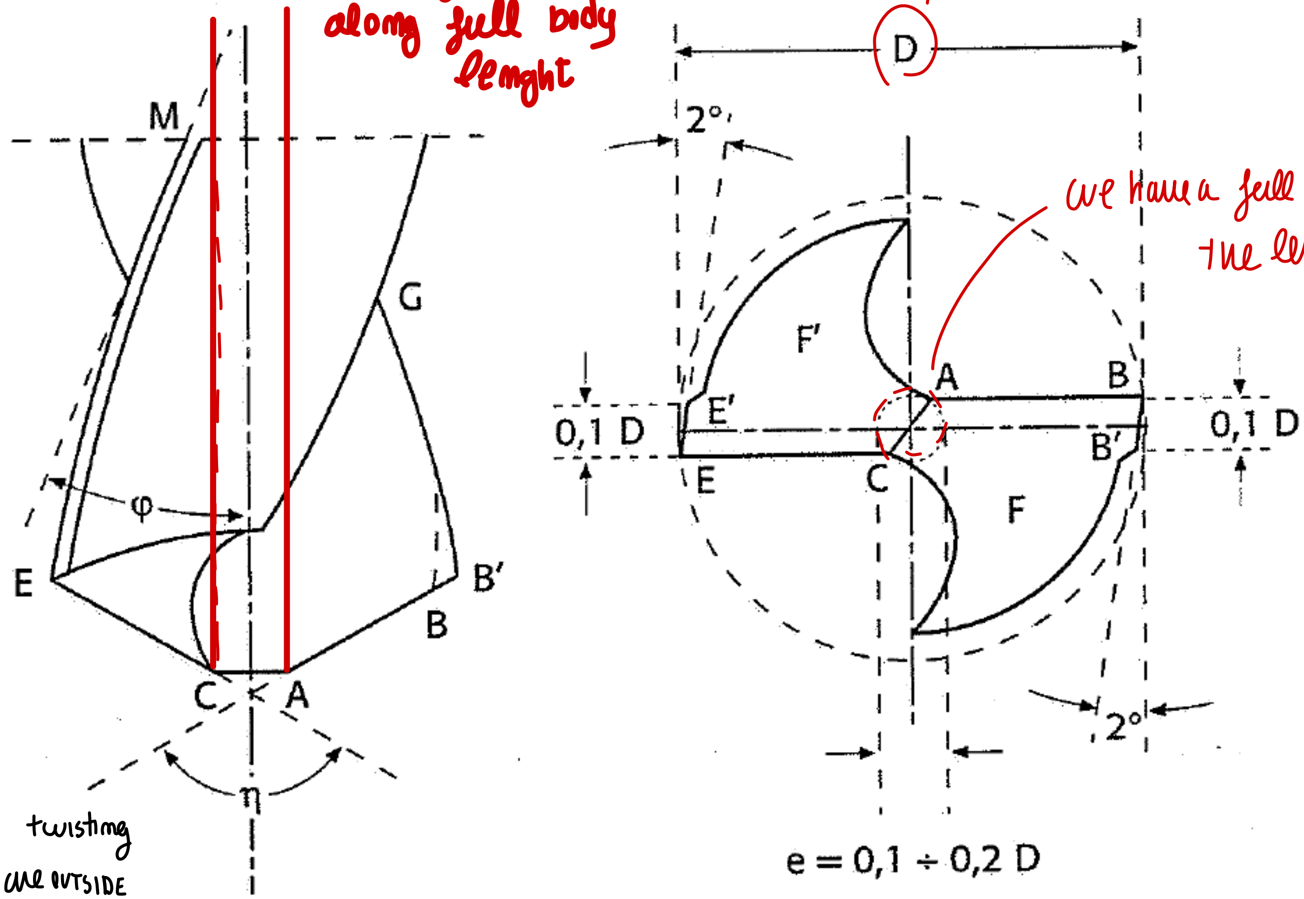
NOT in this way

enable to have more ROBUST tool



to avoid a continuous contact (so reduce friction) between the flute body and the cutted surface
only the MARGIN LIP (*) remain always in contact

Tool: Twist Drill



Full cylinder along full body lenght

tool diameter is related to HOLE to make

we have a full cylinder along the lenght of Body

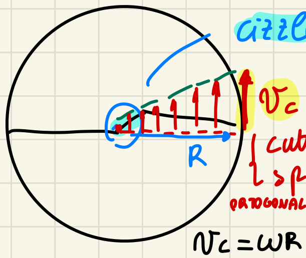
the FLUTES twisting around axis are outside respect cylinder → MORE ROBUST TOOL

(*) look from BOTTOM (why cutting edge don't reach the axis)

BOTTOM VIEW

TOOL ROTATING with a given velocity

ω

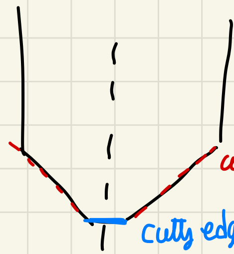


PROCESS parameter

{cutting speed}

ORTOGONAL to Rotation AXIS

side view



$L/D \leq 100$

≤ 100

cutting edges STOP before the axis

cutting edge NOT sharp V but with ⊔ chizzle

here $v_c = 0$

{for many reason we have that CIZZLE}

$v_c = \omega R$

where $R = 0$ (on axis) $\rightarrow v_c = 0$

⇓

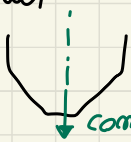
on cizzle, on axis $v_c = 0$

NO CUTTING on the axis

we can't perform cutting there

no material removal,

it is enough to have the CIZZLE ...

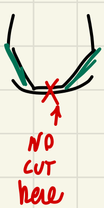


compression on material along axis of motion

use CIZZLE to deform and remove the material along cutting edges ⇐

It is un-usefull to have CUTTING edge on the BOTTOM, ...

We have FLAT ELEMENT



We move material here where we CUT IT

compress material that goes to the CUTTING edges, where it will be CUT

α : CIZZLE edge angle → depending on the material that we are CUTTING

TOOL VIBRATING during cutting, so usually $D_{hole} > D$ → inside tollerance reached

twist drill to make hole and than other works done on it!

$\underbrace{\text{twist drill}}$

↓

limited quality IT10 max ← TOLLERANCE grade reffered to the scale

↑ to have a better quality ≃ IT7 of holes (less variability) you need to perform further operations

HOLE MAKING

twist drill to make hole

+

finish with other tool

enlarge later the hole ←

TOOL: TWIST DRILL

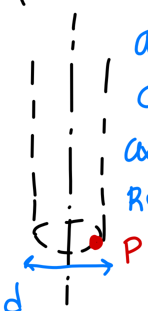
TOOL ANGLES

NORMAL CROSS SECTION angles

⇓

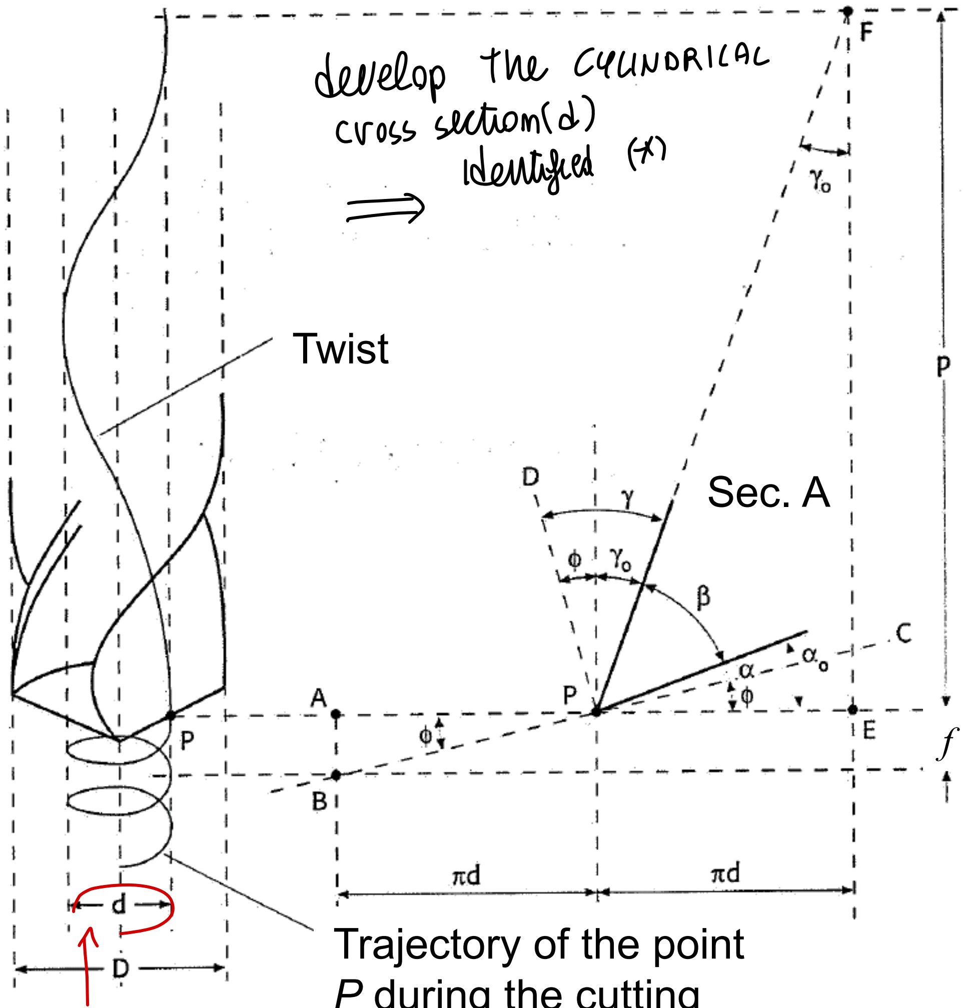
to see that angles, you can consider a point P generic of the cutting edge (with distance d small)

and consider a cylinder with axis as axis of ROTATION



↓

CROSS SECTION of tool by that cylinder



develop the CYLINDRICAL cross section(d) identified (*) ⟹

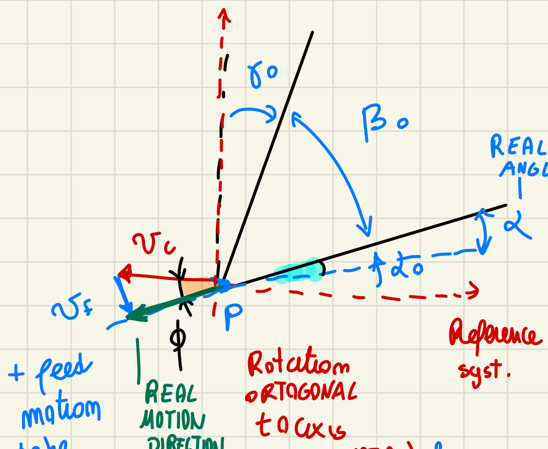
generic point in position d

(*) obtaining a figure similar to TURNING

cylindrical CROSS SECTION

$$\mathrm{tg}\,\phi = \frac{f}{\pi \cdot d}$$

⇒ looking to that geometry:



(CUTTING MOTION) of generic POINT (P) of the CUTTING EDGE

$$\alpha = \alpha_0 - \phi$$

$$\tan\phi = \frac{f}{\pi d}$$

given feed

(full ROTATION in position d)

depend on considered point → RESPECT TURNING here d vary a lot...

$0 \lesssim d < D$ vary to total D from axis mean

the real α vary along the cutting edge from POINT TO POINT

IF d vary on Big Range, also ϕ vary along a Big Range

so $\alpha = \alpha_0 - \phi$

costant ← α_0 costant value of tool

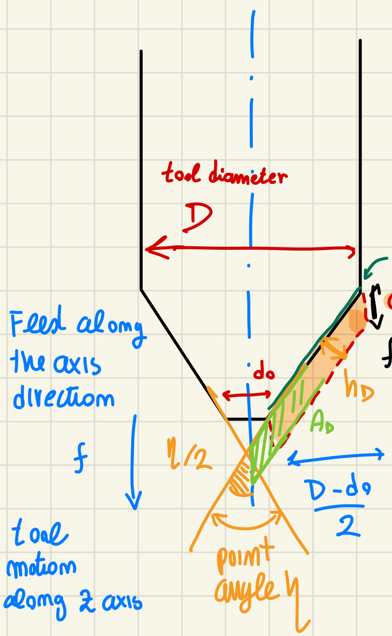
to have α constant, we should shape the tool in a way that α_0 vary in each point along cutting edge → COMPLEX TOOL!

α NOT CONSTANT... to have $\alpha \simeq$ const we have to shape the tool so that α_0 different in each point

COMPLEX shape of tool

looking the TOOL

PROCESS PARAMETER:



used CUTTING EDGE for material cutting

chip before Deformation

Area of chip

theorethical area

$$A_D = \frac{D-d_0}{2}\,\frac{f}{2} \quad (NO)$$

BUT the real area of chip removed A_D is: $D/2 \cdot f/2$

the one consider

$$A_D = \frac{D}{2}\,\frac{f}{2}$$

this is the real one

we consider a longer area down to the axis

thickness (h_D) of chip before formation

$$f_z = \frac{f}{Z} = \frac{f}{2}$$

$$\Rightarrow h_D = \frac{f}{2}\sin\left(\frac{\gamma}{2}\right)$$

because we have 2 CUTTING EDGES in one full ROTATION we moove the tool along feed direction (f)

so the $f_z = f /$ tip number

Z: # of tool tips

here on that drilly $Z = 2$

do: diameter of the CIZZLE, cylinder of cizzle

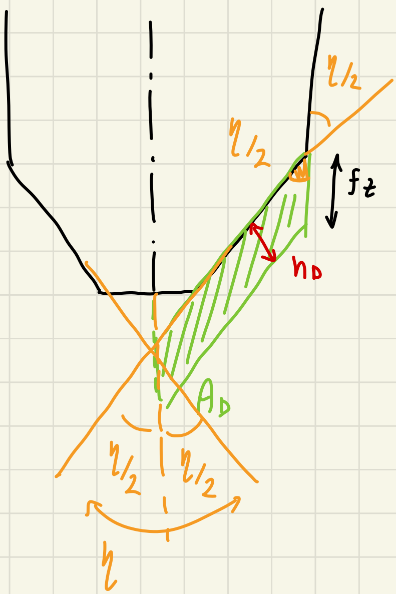
$f/2$ is the real amount of material that a single cutting edge cut (2 cutting edge walking)

We consider all A_D (chip before def.)

Cause CIZZLE deform the material below CUTTING EDGE, also that material is CUT!

matenal behind cluzzle is moored below one CUTTING EDGE.. so it
is CUTTED anyway $A_D = \frac{D}{2} \cdot \frac{f}{2}$ all removed!

$\underbrace{\phantom{A_D = \frac{D}{2}}}$ chip area, full matenal cutted, also the one in front of cizzle



Thickness of chip (h_D)
↑ evaluated ⊥ to CUTTING edge

h_D is related to f_z of the cutting edge
and depends on POINT angle η

$$h_D = f_z \sin\left(\frac{\eta}{2}\right) = \frac{f}{2} \sin\left(\frac{\eta}{2}\right)$$

usefull to predict CUTTING FORCE
thanks to KRONENBERG
Relationship

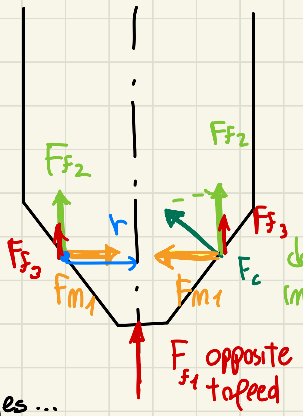
↳ to apply KRONENBERG
and compute F_c ⟹

FORCES

We have multiple FORCES components: {simple drawing for BASIC ELEMENTS}

Side view | **Bottom view**

mainly 3 forces components

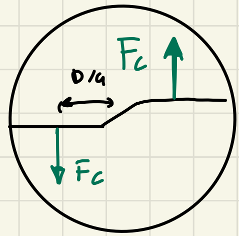


decomposed into

We have 2 cutting edges...

F_{f_1} opposite to feed

(feed force opposition)



parallel to cutting speed direction

+ during chip formation...
force ORTOGONAL to cutting edge

$\Downarrow$

$$F_f = F_{f_1} + 2 F_{f_2} + 2 F_{f_3}$$

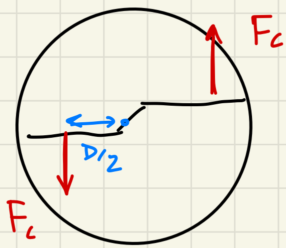
$$r = \frac{D + d_0}{4} \underset{\text{approx}}{\simeq} D/4$$

from...

$$F_c = K_c A_D$$

$$A_D = \frac{D}{2} \frac{f}{2}$$

$$K_c = \frac{K_{cs}}{(h_D)^x}$$



Torque
$\downarrow$

$$T_c = F_c \frac{D/2}{1000} \; [Nm]$$

$$P_c = T_c \cdot \omega \; [W]$$

$$\longrightarrow \quad t_c = \frac{V}{MRR}$$

$$MRR = \frac{\pi}{4} D^2 v_f = \frac{\cancel{\pi}}{4} D^{\cancel{2}} f \frac{v_c}{\cancel{\pi D}}$$

$$v_f = f \cdot n = f \cdot \frac{v_c}{\pi D}$$

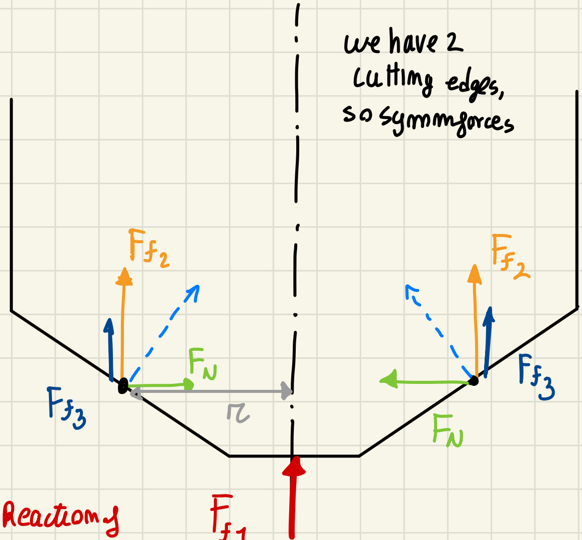
$$\{ v_c = \pi D n \rightarrow n = v_c / (\pi D) \}$$

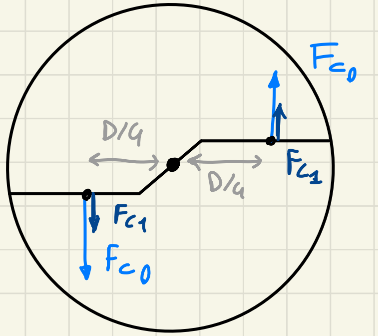
$\Downarrow$

$$MRR = \frac{Df}{4} v_c = A_D \cdot v_c$$

Forces components

we have 2 cutting edges, so symm forces





Reaction of material, due to cizzle deformation

(Force opposite to feed direction, Feed Force 1)

(we'll have many components along feed)

for chip formation, the RESULTANT force stay in a plane ⊥ to cutting edge (3D in space)

(you can decompose resultant)

{ (one component on side view plane)
{ // // bottom view (⊥ PLANE)

(parallel to the cutting speed direction)

decomposed along feed, radius
F_{f_2} F_N

the point where consider the forces, is at radius $r = \frac{D+d_0}{2}$ average $\cong (D/2)\,1/2$ APPROX in the middle

$\cong D/4$ of the tool radius

and we have other components due to FRICTION (friction of chip on Rake face) decomposed on his component

F_{c_1} // to cutting force

F_{f_3} // to feed

so on feed direction we have more forces

$$F_f = F_{f_1} + 2F_{f_2} + 2F_{f_3}$$ Feed direction

$$F_c = F_{c0} + F_{c_1}$$ cutting direction

⇓

POWER: $\cancel{P = \vec{F}\cdot\vec{v}} = \cancel{F_c v_c}$

complex formula, NO!

We prefer to use other formula... to evaluate P

When speaking on DRILLING → on ROTORY tools (tool moving with a certain cutting speed)

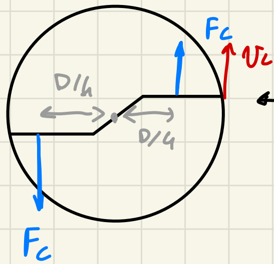
POWER

Is Better to use the TORQUE

being

$$F_c = K_c \cdot A_D$$

$$\begin{cases} A_D = \frac{D}{2} \frac{f}{2} \\ K_c = \frac{K_{cs}}{(h_D)^x} \end{cases}$$



torque applyed by tool

$$T_c = F_c \cdot \frac{D/2}{1000} \quad [N \cdot m]$$

$$\Downarrow$$

$$P_c = T_c \cdot \omega \quad [W]$$

avoid computationg forces F_s, F_c...
because the v_c velocity on ext. diameter
but we develop the force F_c on half of D/2 ..
in the middle of cutting edges

↓

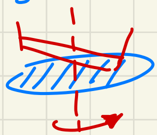
It is better to use the TORQUE for POWER computation

CUTTING TIME to complete the process description

⇓ how to evaluate t_c (more ways!)

$$t_c = \frac{V}{MRR}$$

where on DRILLING: MRR cutting a circle, moving tool in a direction ortogonal to the circle



$$v_f = f \cdot n = f \cdot \frac{v_c}{\pi D}$$

$$v_c = \pi D n \rightarrow n = \frac{v_c}{\pi D}$$

feed vel.

$$MRR = \frac{\pi}{4} D^2 \cdot v_f =$$

$$= \frac{\cancel{\pi}}{4} D^{\cancel{2}} \cdot f \frac{v_c}{\cancel{\pi} \cancel{D}} \rightarrow MRR = \frac{D f}{4} v_c = A_D v_c$$

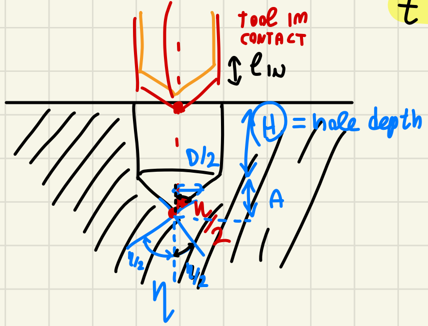
A_D

same expression of MRR like ORTOG. CUTTING and TURNING!

the issue here is that t_c is an underestimation of the time

↳ if we look to the full motion

supposing FULL MOTION BLIND hole situation



$$t_c \simeq \frac{\overbrace{H + A}^{\text{all tool moovement while contact}}}{v_f}$$

correct way to compute the time

(A is related to TIP angle η)

$$\tan\left(\frac{\eta}{2}\right) = \frac{D/2}{A}$$
$$\Downarrow$$
$$A = \frac{D/2}{\tan(\eta/2)}$$

Knowing hole depth H, and diameter D, and shape of tool η → we can estimate A easily and t_c from here...

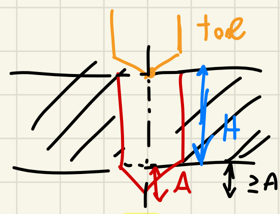
(all process time)

machining time → extra moovement... due to an extra path IN (ℓ_{IN}) of tool motion (safe position)

$$t_m \simeq \frac{\overbrace{H + A}^{\text{CONTACT}} + \overbrace{\ell_{IN}}^{\text{EXTRA PATH}}}{v_f}$$

APPROX.. it miss something!

on a through hole → the conical portion of tool should exit on the otherside if through hole...



for Through hole

$$t_{contact} = \frac{H + A}{v_f}$$

while all machining should be:

$$t_m^* \simeq \frac{H + A + \ell_{IN} + \overbrace{\ell_{OUT}}^{\text{extra out part}}}{v_f}$$

NO

NOT correct, something still missing!

→ we are ENTERING and EXITING from the hole {needed exit motion}

IF MOVING with same velocity

(sometimes the motion of exit the hole can be done with same process parameter and have $t_m = 2t_m^*$)

$$\downarrow$$

$$t_m \simeq 2\,\frac{H + A + \ell_{IN} + \ell_{OUT}}{v_f}$$

Usually the removal is faster!

↙ so the expression changes

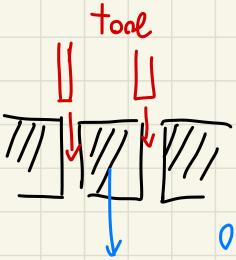
you can exit the hole in RAPID MOTION! small time

Return / so on exit you have $\dfrac{H + A + \ell_{IN} + \ell_{OUT}}{v_{RAPID}} = t_{RETURN}$

$\rightarrow t = t_{RETURN} + t_m$

We have to discuss **HOLE MAKING** in a different situation

through hole creation made by simply a RING tool, NOT cutting the full material but the perimeter



ONLY for THROUGH HOLES

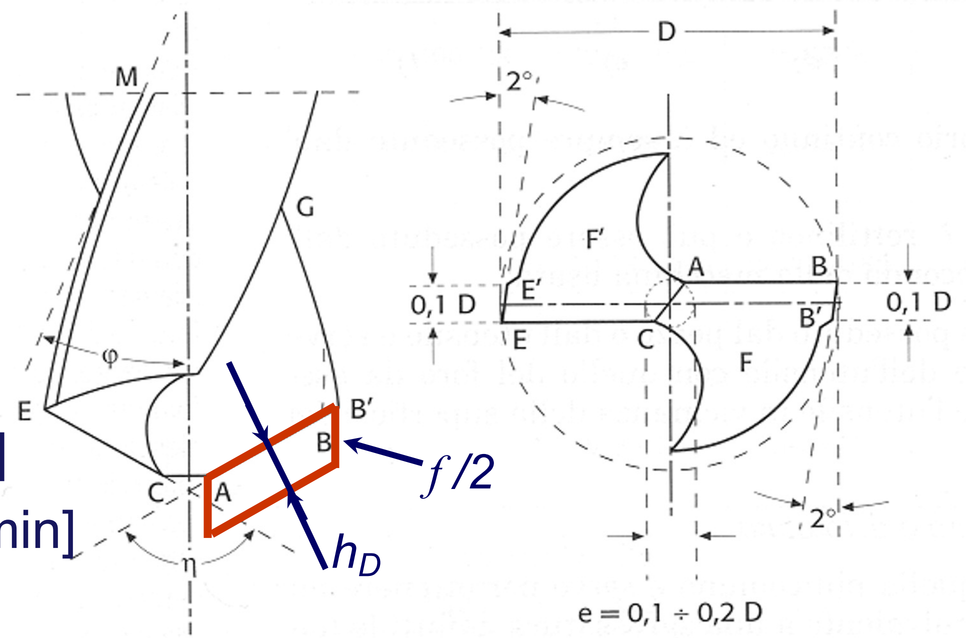
- efficient cutting for BIG HOLES

instead of remove all you just remove a small portion and a cilinder goes away

DRILLING

Characteristic cutting parameters:

- Feed per revolution f [mm/rev]
- Drill diameter D [mm]
- Chip thickness h_D [mm]
- Chip section $A_D = f D/4$ [mm^2]
- Rotational speed of the drill n [rpm]
- Cutting speed $v_c = \pi\, n\, D/1000$ [m/min]



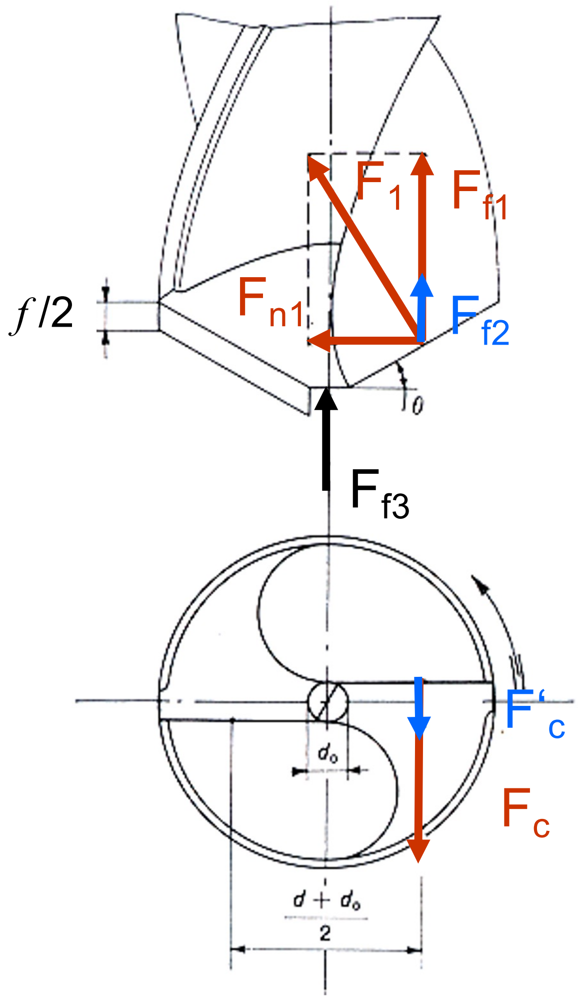
Range:

f = 0,01 ÷ 0,8 mm/rev

D = 1 ÷ 50 mm

v_c = 5 ÷ 300 m/min

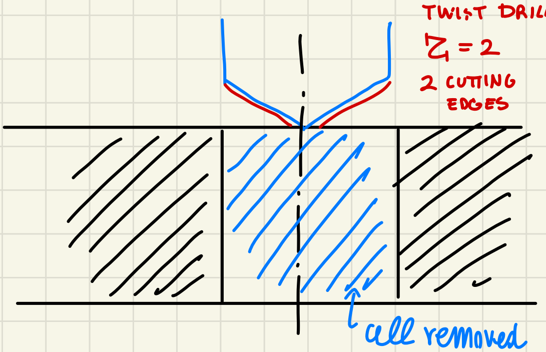
Cutting Forces in Drilling



1 Forces due to chip formation

2 Forces due to friction on flank faces

3 Force on chisel edge due to feed



TWIST DRILLING

$Z = 2$

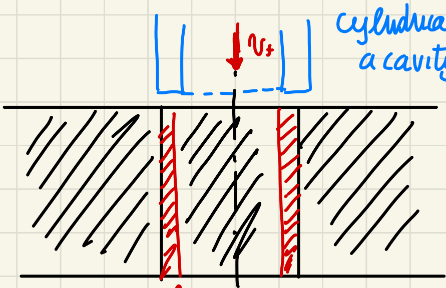
2 CUTTING EDGES

Through hole

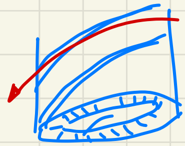
Z = theeth of the tool

IF made by the classical TWIST DRILL to make the hole

instead to reduce the needed work for material removal → we remove JUST a small portion TRAPANING



cylindrical tool with a cavity



tool

feed motion

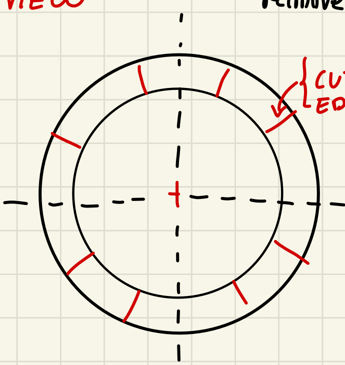
$v_f = f \cdot n$

$Z \geq 2$

↓

multiple cutting edge cutting simultaneusly

TOP VIEW



all CUTTING EDGES WORKING simultaneusly during full rotation (like TWIST DRILLING)

remove a Ring of material

↓ for this kind of cutting you use a MULTI CUTTING EDGE TOOL

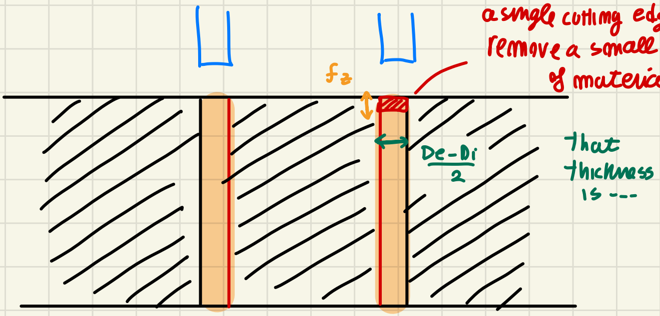
if $v_f = f \cdot n$

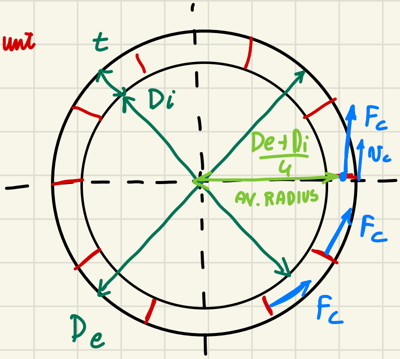
IF you move along the axis of a feed total of (f) displacement

→ a single cutting edge remove an amount of material equal to

⇐ $f_z = f / Z$

here there are difference from twist drilling





$A_D = f_z \cdot \frac{De - Di}{2}$

∀ cutting edge..
F_c is normal to rt..
tangential respect circular CUTTING MOTION
(F_c in the middle of cut edge, v_c in the perimeter)

$t = \frac{De - Di}{2}$

$F_c = K_c \cdot A_D$ cutting pressure method
(same ∀ edge)

T_c: total work related to Z_i acting CUTTING EDGES
sum of all components F_c
↳

$$\boxed{T_c = Z_i\, F_c\, \frac{De + Di}{4} \frac{1}{1000}} \quad [N \cdot m]$$

$P_c = T_c \cdot \omega$ where $\omega = \frac{2\pi n}{60}$

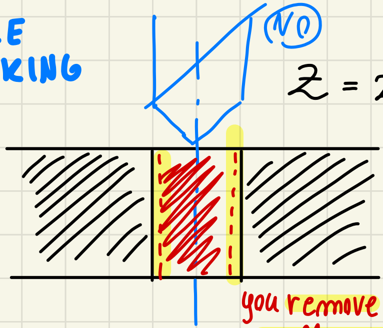
removing the external portion of the hole.. small amount of material

So less total energy needed, comparing full removal respect just trapaning → less energy

only if you machine a THROUGH HOLE, where inside material is removed ← [EXTERNAL RING removal] (NOT for BLIND HOLE)

HOLE MAKING

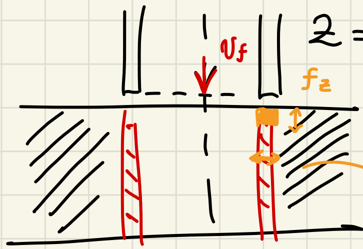
THROUGH HOLE



$z = 2$

you remove small portion around the big hole to make.

$\Longrightarrow$



$z = 2$

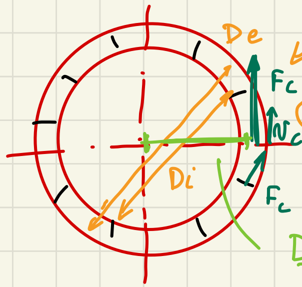
$v_f = f \cdot n$

you need a tool to remove the material.

all cutting edges working simultaneously...

$$f_z = \frac{f}{Z}$$

(ABOVE POV)



thickness removed $(De - Di)/2$

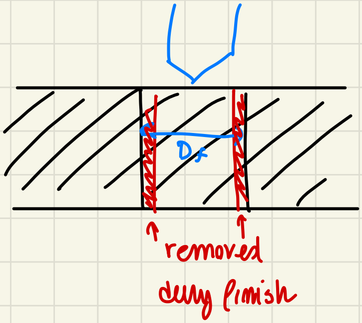
$$A_D = f_z \cdot \frac{De - Di}{2}$$

$$F_c = K_c \cdot A_D$$

$$P_c = T_c \cdot \omega \Leftarrow T_c = z \cdot F_c \cdot \frac{De + Di}{4} \left(\frac{1}{1000} \right) [N \cdot m]$$

$$\omega = \frac{2 \pi n}{60}$$

Roughing / Finish operation : MAKING HOLES

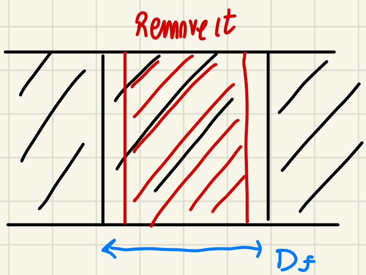


you leave
machining allowance,
use TOOL to enlarge
the hole
similar situation below.

RIMMING operation
multi edge cutting

When MAKING HOLES... we may have : • ROUGHING
• FINISHING operations
enable to obtain better accuracy in Roughness and size/shape of hole

When hole



final goal Df

WITH DRILL WE JUST
remove a certain portion

leaving machining
allowance -> material removed during FINISHING

TWIST DRILL + enlarging hole tool

during FINISH,
same computation
approach -> removing remaining portion
⇐ FINISHING made by a similar tool,
We compute @ same way the F_c, P_c, ---

enlarge holes → HOLE MAKING ≡ REAMING (FINISH) operation (Multi cutting - edges tool)

similar to TRAPANING

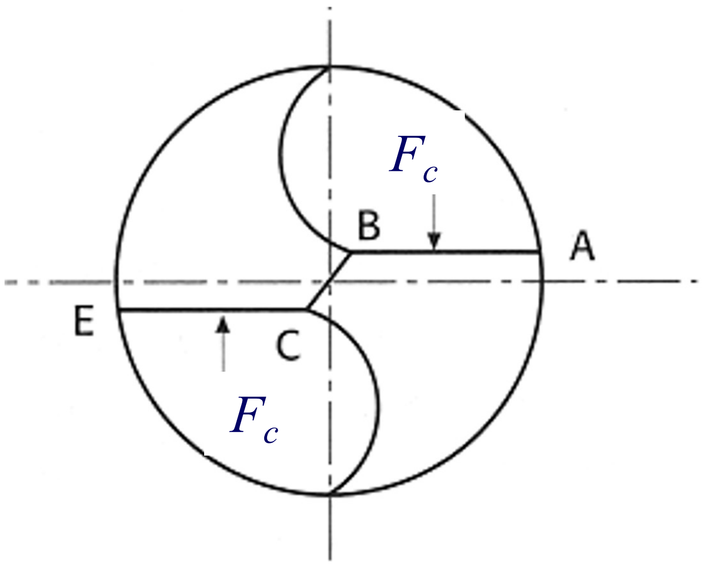
Cutting Forces in Drilling

Even in drilling we can use the specific pressure method.

Considered the chip section A_D,

the F_c force is given by:

$$F_c = k_c A_D \quad [\mathrm{N}]$$



and then the cutting torque is given by

$$T_c = \frac{F_c \cdot D/2}{1000} = \frac{k_c A_D \cdot D}{2 \cdot 1000} \quad [\mathrm{Nm}]$$

CUTTING POWER IN DRILLING

The cutting power in drilling is given by:

$$P_c = T_c \cdot \omega \quad [\mathrm{W}]$$

since:

$$\omega = \frac{2 \cdot \pi \cdot n}{60}$$

it follows that:

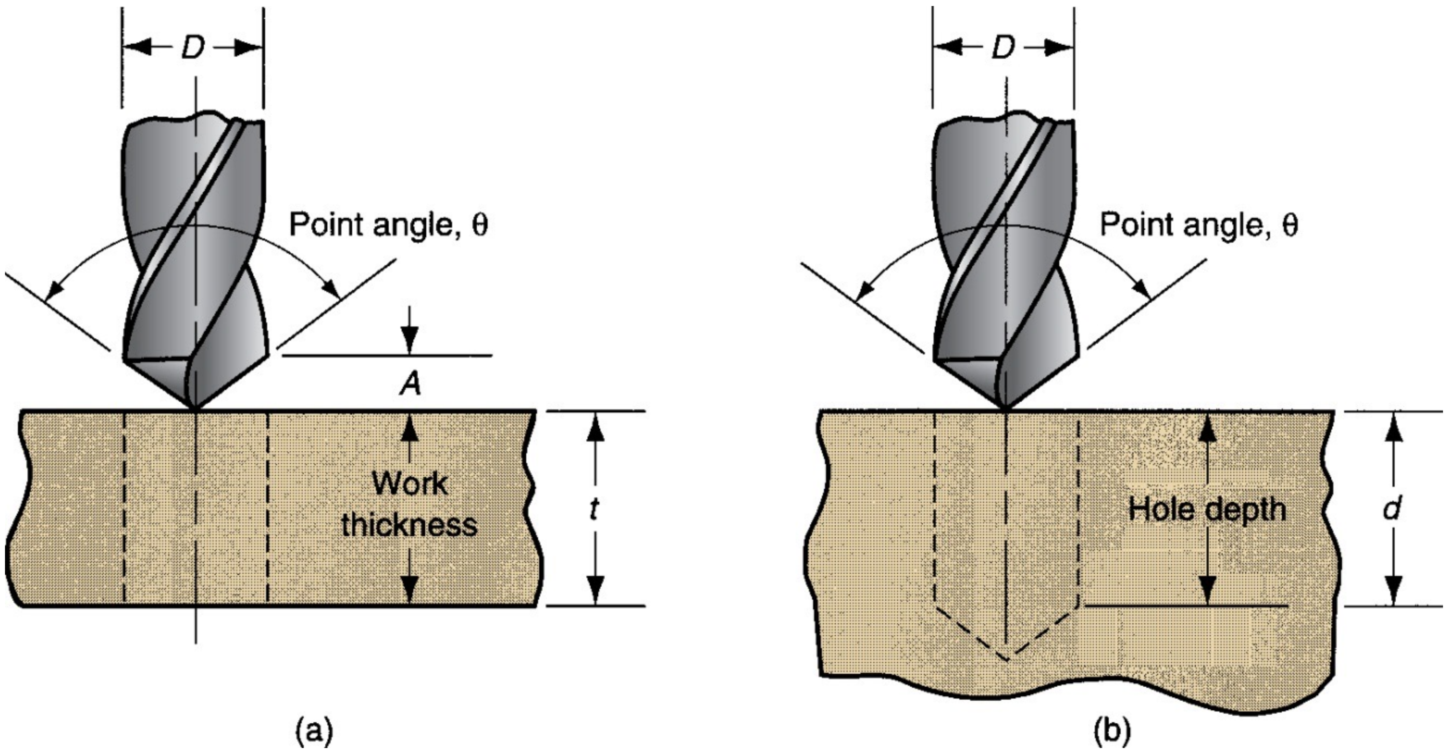
$$P_c = \frac{T_c \cdot \omega}{1000} = \frac{k_c A_D D \cdot \pi \cdot n}{60 \cdot 10^6} \quad [\mathrm{kW}]$$

Drilling Time

Cutting time:

$$t_c = \frac{L + A}{v_f} = \frac{L + A}{Z f_Z n}$$

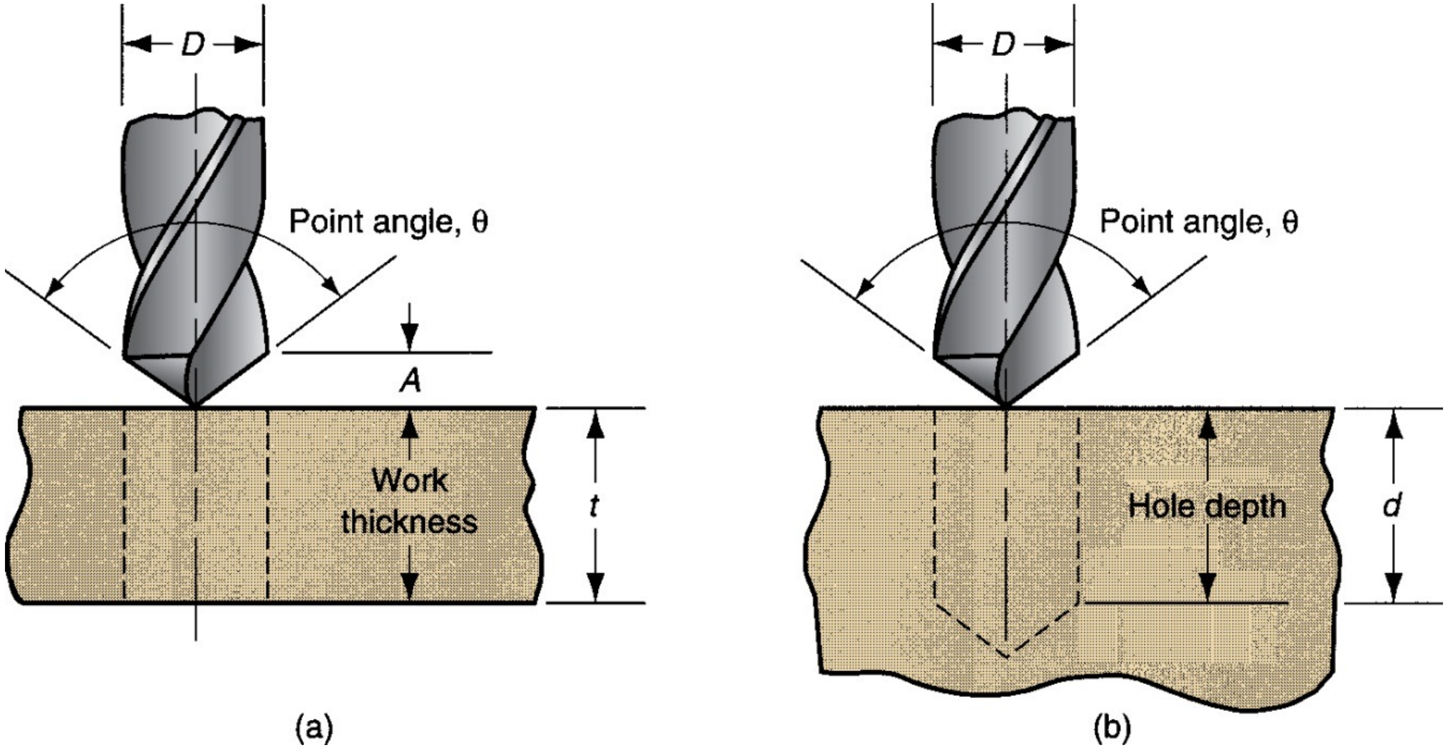
Where L is the workpiece thickness t (a - through hole) or the hole depth d (b - blind hole).



Drilling Time

Cutting time:

$$t_c = \frac{V}{MRR} \qquad MRR = \frac{\pi}{4} D^2 v_f$$

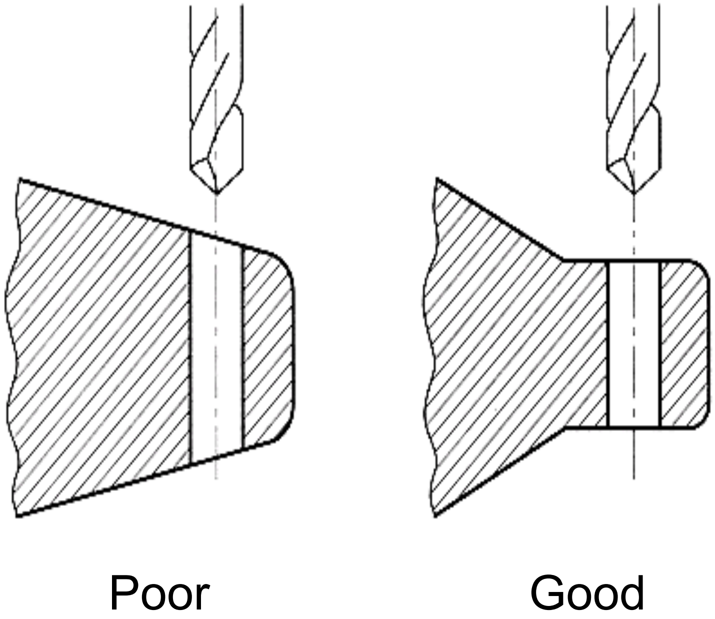


DRILLING OPERATIONS

Often a critical problem in drilling is the tight tolerances of the hole.

Some basic rules:

- generate a flat entering surface, normal to the drill axis, to avoid bending of the drill;
- drill the hole in many passes, with a progressive increase of the drill hole.



Operations Related to Drilling

OPERATIONS that we can perform when DRILLING (making holes)

(a) Reaming, (b) Tapping, (c) Counterboring

FINISHING

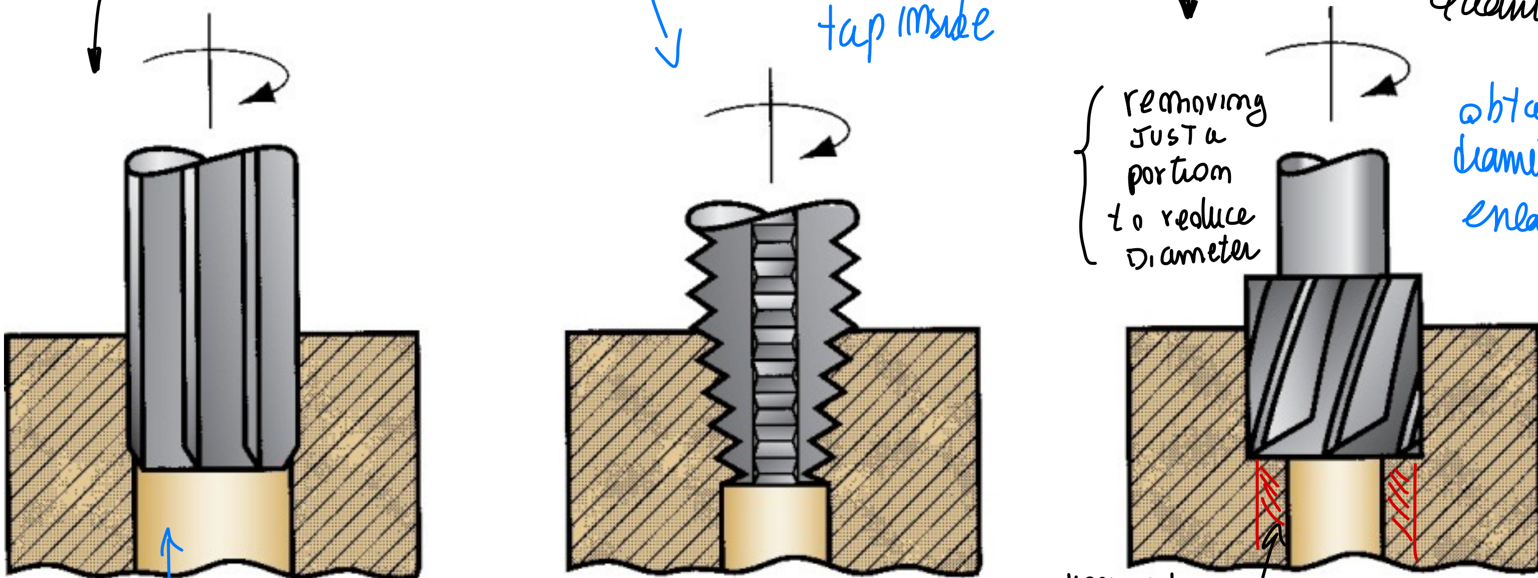
tool to make tap inside

ENLARGE DIAMETER

remove LARGE quantity!

removing just a portion to reduce Diameter

obtain different diameter cylinder, enlarge the hole!



MULTI-CUTTING edge

removed RING material

remove a portion

(a) (b) (c)

here you remove a small portion of material → accurate holes

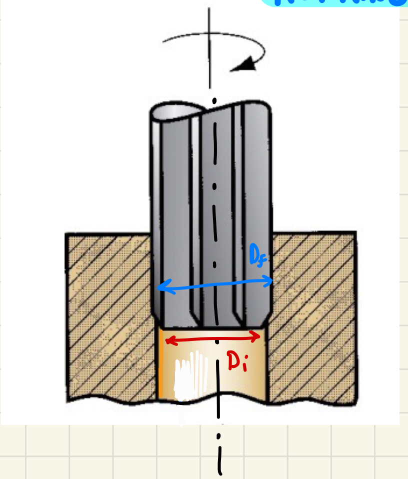
to make threads inside the hole (TAPPING)

tool to enlarge the hole

remove all the material

ROUGHING operation

REAMING



removing a small amount of material on the side of the hole
↓
depth of cut $a_p = D_f - D_i$

Notice: the choise of tool and machining step depends on the feature to create → PROCESS PLANNING

sequence of STEP to create your feature as desired with needed accuracy

COUNTERBORING → removing material to enlarge the hole → ENLARGE Holes
(ROUGHING → scossatura)

VS

while REAMING → for accurate holes → make precision holes

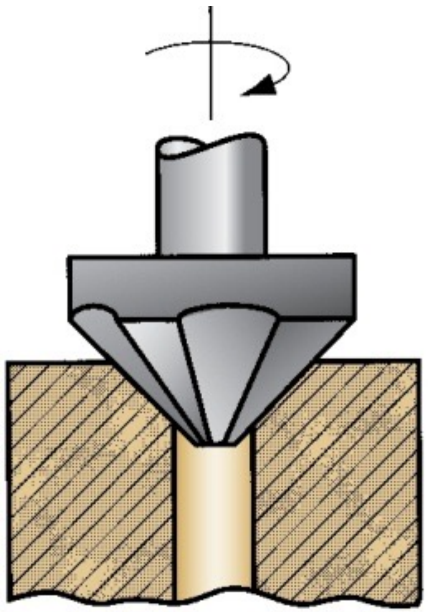
More Operations Related to Drilling

others hole making operations

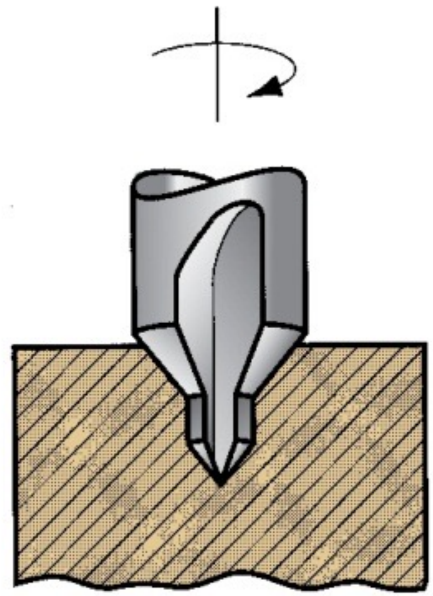
(d) Countersinking, (e) Center drilling, (f) Spot facing

conical shape hole (multi-cutting edge)

to create a flat surface respect which we can move ortogonally the TWIST DRILL

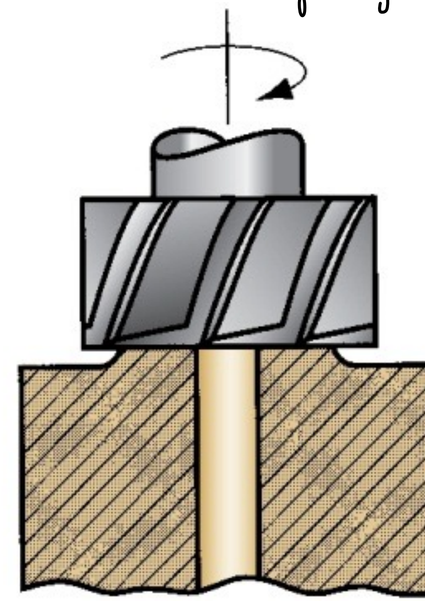


(d)



(e)

used to obtain a good hole guide an have accuracy



(f)

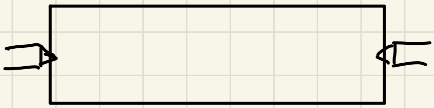
(simulants MILLING)

particular tool to make holes already considered

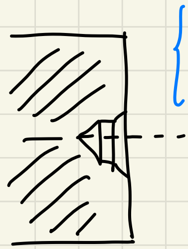
When speaking of turning and FIXTURING...

to FIX we need a conical like shape

We can have
Between center situation →



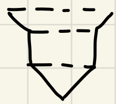
centering hole to
guide the twist drill
inside the hole



{ center drilling (e) }

to obtain that
hole shape in
one operation

conical hole
cylinder
cone



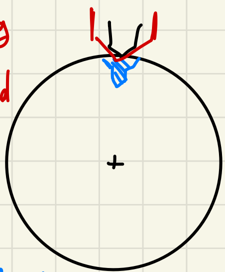
one machining to create the
hole needed for centering

Center the WORKPIECE
Respect machine.. allign the
axis of spindle with PART ROTATION

(IF NOT a flat surface)

If you have an hole on a complex surface

big
tool
needed



part fixed
properly

using twist drill

(Risk operation)

RADIAL Hole → if using twist drill

You have an issue of the HOLDING effect,
the parts tend to go away
and DRILL Bend! PROBLEM! RISK!

IF perform small hole, conical
shape small to define the axis
of the hole → CENTERING HOLE!

to have a better
DRILLING operation!

center
drilling
↓
twist
drilling → other
process

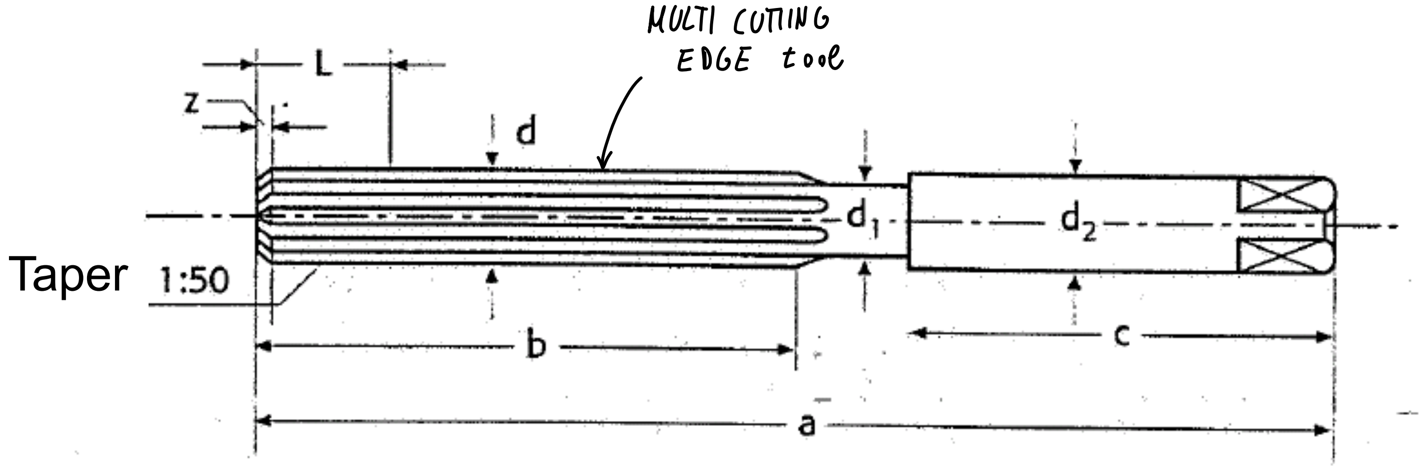
IMPROVE QUALITY
of result → good hole position

REAMING TOOL

multi cutting edge →

When assigned tolerances are very tight, we can use particular tool: a reamer. The operation is called reaming.

The reaming tools do not drill new holes, they only enlarge existing holes (they are slightly tapered).



all cutting edges removing cutting

straight cutting edges // to axis (here)

we may have also helicoidal shape with an angle on the CUTTING EDGE ⇒

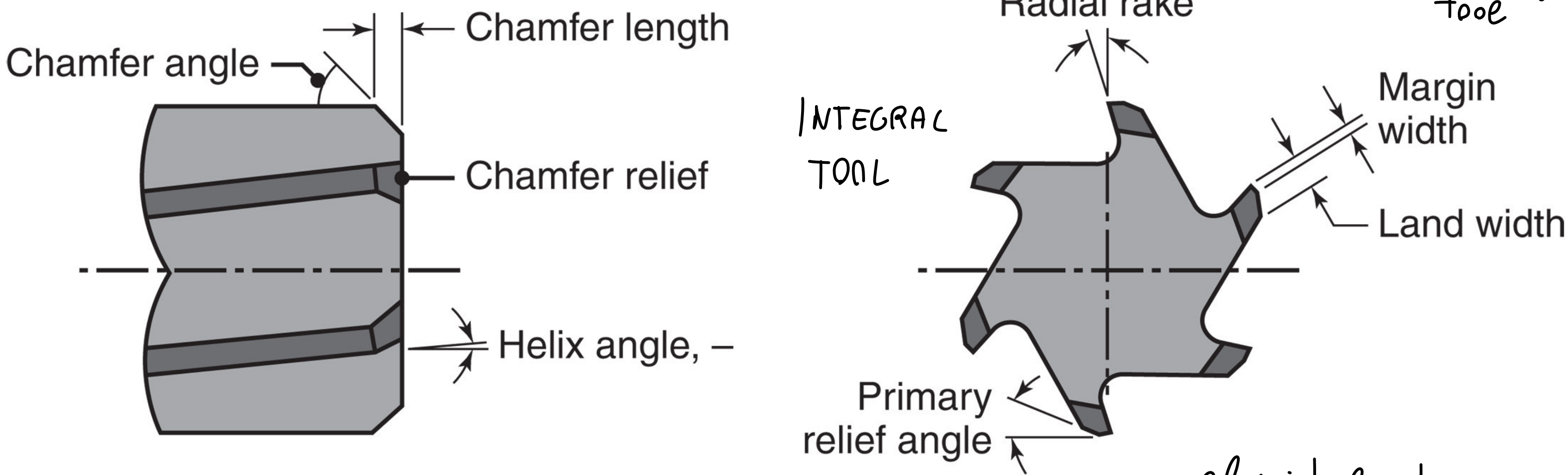
REAMERS

IF with angle of the cutting edge, helicoidal shape

simultaneously finishing the hole

Z = 6 cutting edges tool

INTEGRAL TOOL

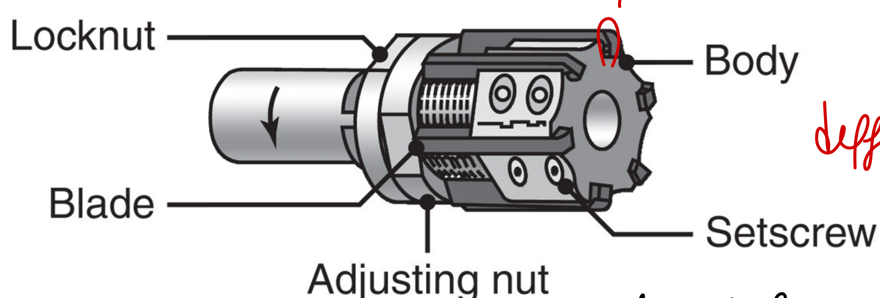


(a)

elicoidal shape of the drill-- radial tools cutting edge

more complex tool

multiple CUTTING EDGE



(b)

different element cutting edge

You can mechanically control CUTTING edge position

set up tool with right Diameter → CORRECT HOLE

REAMING VS BORING

Internal cylindrical machining finishing
(finish the hole) → INTERNAL CYL. TURNING with a single / multiple CUTTING edge tool

shank different respect the one used for external maching because it has to enter the hole and finish it

depending on how much material to remove we may have same accuracy of Reamer, or less accuracy like enlarging hole

---

When making an Hole → TWIST DRILL make an hole with given D

- IF consider an INTEGRAL RIMMER like (a) you can obtain hole with fix D final
- IF change D you need new tool

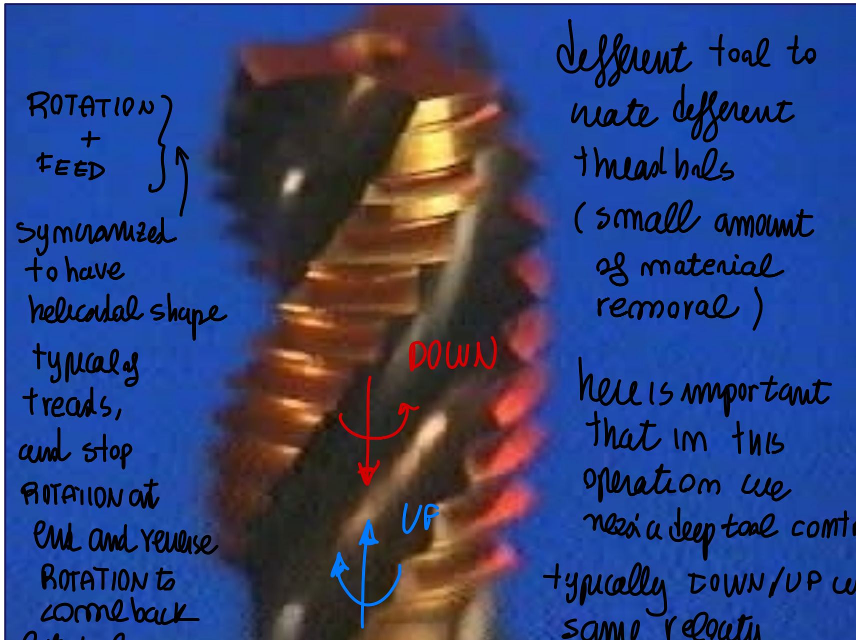
OR adjust the Diam. like (b)

BORING OPERATION → done by a single CUTTING edge ... when MULTI cutting edges (finishing) → we speak about REAMING

TAPPING AND TAPS

Multi cutting edge

particular shape respect thread standard! TOOL for a given thread



ROTATION + FEED

syncronized to have helicoidal shape typical of treads, and stop ROTATION at end and reverse ROTATION to come back let tool out

DOWN

UP

different tool to create different thread holes (small amount of material removal)

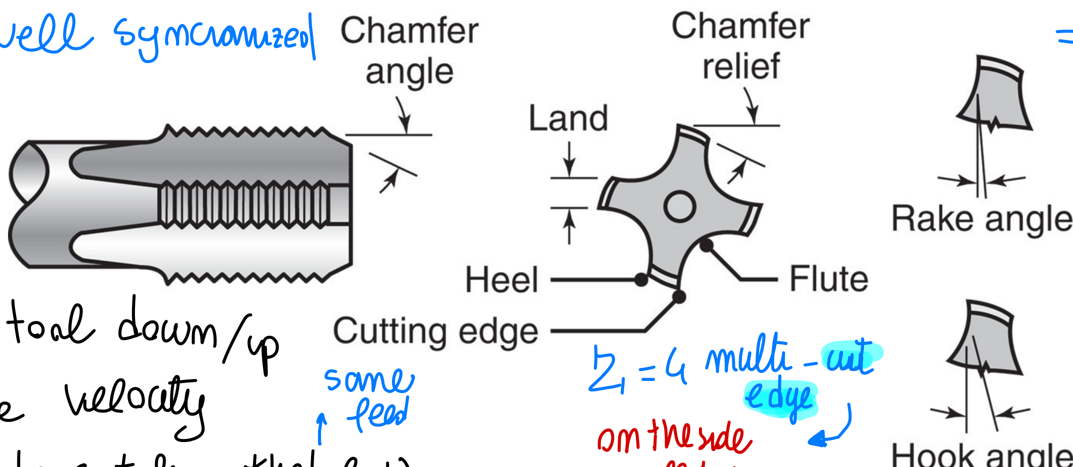
here is important that in this operation we need a deep tool control

typically DOWN/UP with same velocity

complete HOLE MAKING

• TWIST DRILL for hole

+ TAP for tapping making



Rotation + Feed motion — well syncronized

move the tool down/up with same velocity (same time to exit from the hole!)

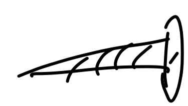
same feed

$Z_i = 4$ multi-cut edge

on the side parallel to axis

⇒ TAPPING

Goal is create the typical cavity of tape (thread)



Machine Tool: Drill Press

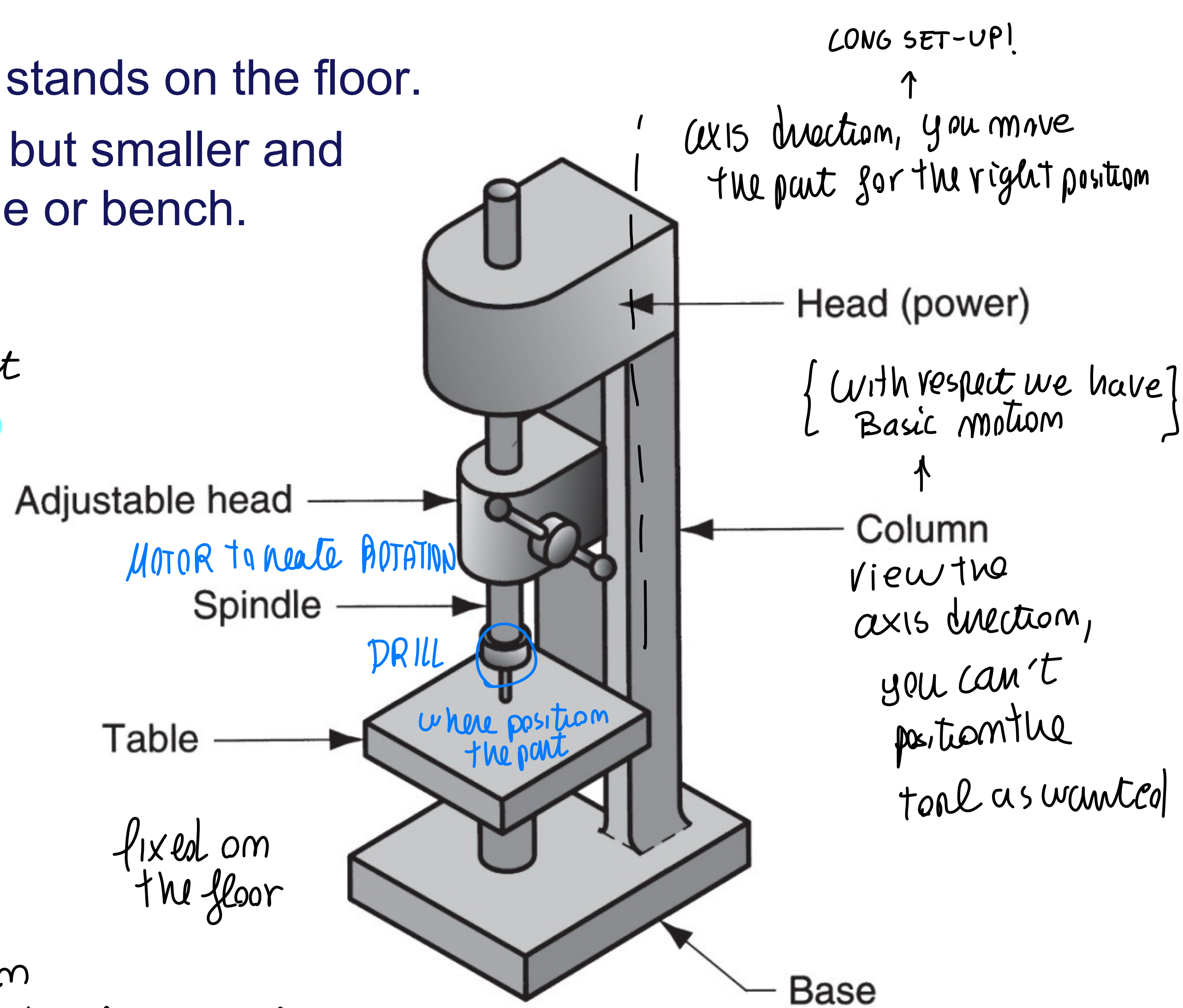
Upright drill press stands on the floor.

Bench drill similar but smaller and mounted on a table or bench.

dedicated machine tool to make that operations of DRILLING

move
- tool respect the part
- part respect tool

↑ Relative motion!



JUST position on the bench → Bench DRILLING

MACHINE TOOL: RADIAL DRILL PRESS

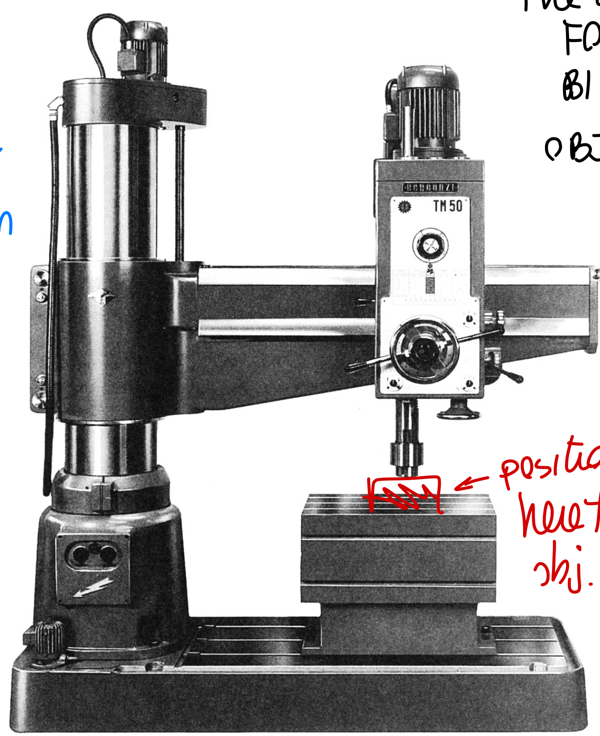
for LARGE PART to drill

position the tool using column motion + ROTATION to radially position the tool

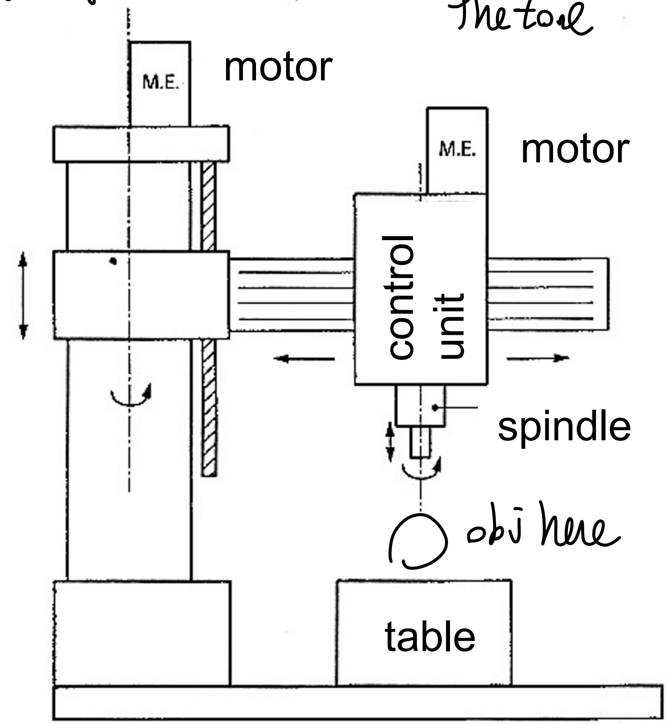
Large drill press designed for large parts

You can position the tool with freedom! FOR BIG OBJECT

position the tool up & down by column



position here the obj.



obj here

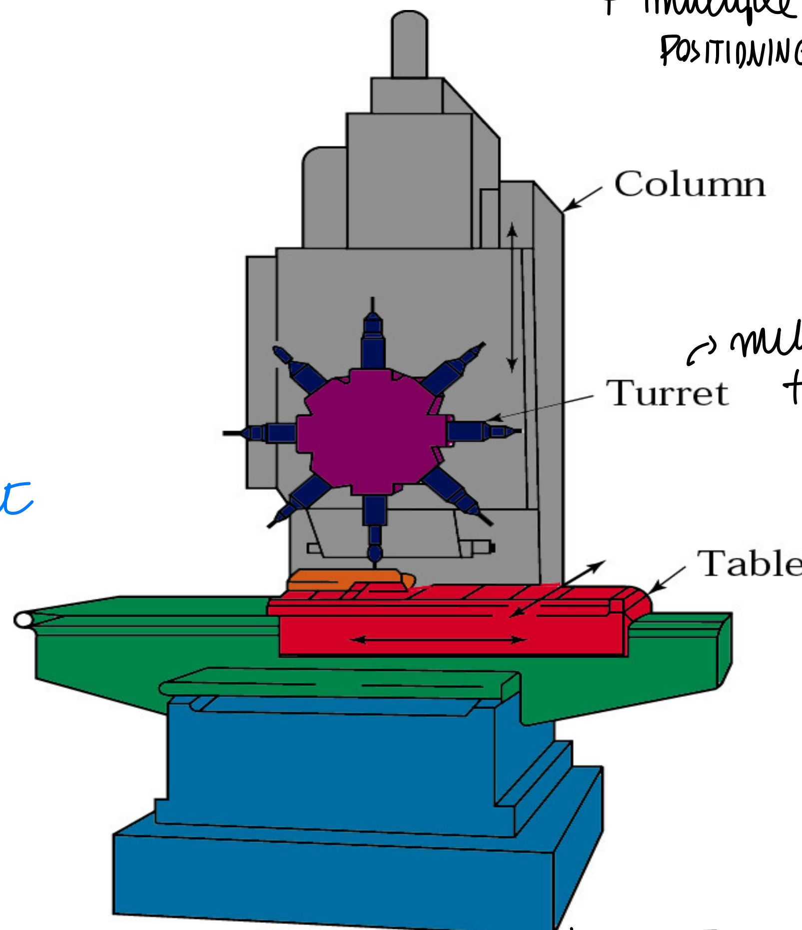
MACHINE TOOL: DRILL PRESS

Dedicated machine to drilling, NUMERICAL CONTROL

Three axis CNC drilling machine

+ multiple motion for POSITIONING the part correctly

dedicated machine used on lines of production (seq. of machine)
⇓
machining line to have higher throughput



→ multiple different tools for hole making

Machining line
used in machining line → each station for one operation!
⇓
HIGH THROUGHPUT

CORRECTLY position the part! ⇒ computer set-up automatic

ROTATE tool and create HOLES and other features, for more machining

WORK HOLDING FOR DRILL PRESSES

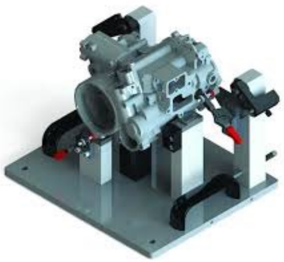
TO IMPROVE DRILLING -- Relevant FIXTURING!

Workpiece in drilling can be clamped in any of the following workholders:

- *Vise* - general purpose workholder with two jaws → planar faces, keeping the part fixed, BLOCK prismatic elements

- *Fixture* - workholding device that is usually custom-designed for the particular workpiece

can be dedicated to a specific shape tool



- *Drill jig* – similar to fixture but also provides a means of guiding the tool during drilling elements that guide the tool, keep axis in right position

more costly set-up... ←

fix element to position

- **DRILL JIG** elements that GUIDE the tool, keeping the axis on wright position → COSTLY SET-UP
but high accuracy
fixture element positioned for perform DRILLING, keeping the tool parallel to desired axis (HIGH ACCURACY)

- **FIXTURE** WORKHOLDING devices in DRILLING/MILLING
like general geometric elements...
(like the VISE two simple planar element blocking a prismatic element)
OR modular element (similar to CMM Fixing)
↑ to create dedicated fix for my application

fixturing devise reshaped depending on the usage
OR if economically convenient → Dedicated fixturing for a specific part HOLDING
(without interfering with the elements)